Supplementary Material for

Life-Cycle Planning of Battery Energy Storage in Distribution Networks With Coupled Electrothermal–Aging Dynamics

This supplement specifies candidate generation and screening, physical execution of daily dispatch, and lifecycle cost. It also reports search coverage, chronological completion status, and selected-day numerical checks. Configuration IDs P01–P11 and model labels M0/M1/M1s/M2 follow the main manuscript.

S1 Candidate generation and screening

S1.1 Design representation

A configuration is $X = \{(b_i, n_i, \rho_i)\}_{i=1}^{K}$, where b_i is a bus, n_i its module count, and ρ_i its power-to-energy ratio. Each module contains 1212 cells of 2.5 Ah and 3.3 V, giving $E_{\text{mod}} = 9.999$ kWh. Thus $E_i = n_i E_{\text{mod}}$, $\bar{P}_i = \rho_i E_i$, and $\bar{S}_i = 1.1\bar{P}_i$. Positive battery active power denotes charging or network consumption.

M0 denotes constant-efficiency operation without aging, M1 the ambient-temperature electrical/aging model, and M2 the dynamic electrothermal/aging model. These three models nominate candidates. M1s is an additional steady-temperature ablation. Electrical and aging equations follow the adopted model of Li et al. [1].

The daily convex objective determines dispatch targets. The short-sequence score below ranks configurations for nomination; full-horizon net present cost (NPC) then ranks the admitted configurations using chronological operation and replacement events.

S1.2 Sampled sequence and annualization

The search space comprises all 59 non-slack buses, at most five stations, and the catalogs

$$\mathcal{M} = \{1, 2, 4, 6, 8, 16, 32\}, \qquad \mathcal{R} = \{0.25, 0.375, 0.5\} \text{ kW/kWh}.$$

Module counts and PCS ratios are selected independently for each installed station.

The sampled sequence combines four three-day seasonal blocks with the days containing the maximum hourly temperature, minimum hourly temperature, maximum hourly radiation, and maximum hourly load. Duplicates are removed and days sorted. The archived sequence is

$$\mathcal{D} = (1, 2, 3, 91, 92, 93, 133, 158, 181, 182, 183, 227, 271, 272, 273, 365). \tag{S1}$$

Each retained day is simulated for 24 hours consecutively in this order. States are carried forward between retained days, subject to the daily SOC rule. Battery age advances only over these 384 simulated hours; omitted calendar intervals do not contribute aging exposure.

Each day has an equal annualization weight of $365/16 = 22.8125$, including the extreme days. These uniform weights give a stress-inclusive screening estimate; their correspondence to annual day frequencies has not been calibrated.

S1.3 The implemented screening score

Let $C^{\mathrm{op}}_{m,d}$ denote simulated operating cost and $A^{\mathrm{proxy}}_{m,d}$ the evaluated monetary aging diagnostic on sampled day d. Nomination uses

$$\widetilde{J}_m(X) = C^0(X) + a(H_s, r)\left[\frac{365}{16}\sum_{d\in\mathcal{D}}(C^{\mathrm{op}}_{m,d} + A^{\mathrm{proxy}}_{m,d}) + C^{\mathrm{OM}}(X)\right], \tag{S2}$$

$$a(H_s, r) = \sum_{y=1}^{H_s}(1+r)^{-y}, \qquad H_s = 10, \quad r = 0.05. \tag{S3}$$

C^0 includes equipment and initial commissioning; baseline O&M is zero. For M0, the aging diagnostic is zero. For M1/M2, the diagnostic is accumulated from the executed trajectory, rather than copied from the convex dispatch objective. Its hourly cell-based structure is

$$A^{\mathrm{proxy}}_{i,h} = \frac{N_i}{2424}\frac{c^{\mathrm{proxy}}}{1800}\alpha_D^2\frac{C^{\mathrm{ref}}}{0.04}\int_h^{h+1}\left[k_{\mathrm{cal}} + (k_{\mathrm{hot}} + k_{\mathrm{cold}})\sqrt{|I|}\right]^2 d\tau_h, \tag{S4}$$

Here N_i is the station cell count and the reference replacement cost is

$$C^{\mathrm{ref}} = 2424(2.5)(3.3)(1800)/1000 \text{ CNY}.$$

The baseline uses $c^{\mathrm{proxy}} = 1800$ CNY/kWh and $\alpha_D = 1$. Stress coefficients are evaluated along the physical trajectory. Cold/high-SOC charging damage enters the physical health update but not this diagnostic.

Equation (S2) approximates long-run wear through an annualized aging penalty. Explicit replacement dates, future purchases, and terminal recovery enter only the subsequent chronological evaluation. Failed or incomplete screens receive no finite score. Screening can therefore exclude a configuration that would perform well under full-horizon accounting.

S1.4 Seeds, proposals, updates, and stopping

The following deterministic procedure is applied independently under M0, M1, and M2.

1. Evaluate 75 seeds: the empty configuration; one station with two modules and $\rho = 0.5$ at each of the 59 buses; and 15 uniform five-site designs at PV buses $\{32, 42, 45, 47, 48\}$, using $n \in \{1, 2, 8, 16, 32\}$ and all three ratios.

2. Start trajectories from three PV-site seeds $(n, \rho) = (2, 0.5), (1, 0.25), (32, 0.25)$, the two lowest-scoring single-site seeds, and the empty design. Remove duplicates.

3. Generate one-station edits: add a station at any unoccupied admissible bus with any catalog specification; remove a station; relocate one station to any unoccupied admissible bus; or replace one station's module count and/or ratio by any catalog pair. Changes need not be adjacent catalog steps.

4. Group proposals as add, remove, relocate, capacity, and PCS edits. Joint module/PCS changes belong to the capacity group. Within each group, prioritize destination buses by single-site seed rank, then by the SHA-256 digest of the current and proposed portfolio keys. Interleave the five groups in this order, taking at most 64 unseen proposals per current design. Interleave across current designs under the remaining shared budget.

5. Cache finite-score and failed evaluations. Update each current design to the lowest-score member of itself and its already evaluated one-edit neighbors; break ties by portfolio key. Equal-score moves are possible. There is no strict-improvement stopping test.

6. Stop the main stage after at most four rounds, 900 distinct evaluations, or an empty proposal batch. Around each of the ten best designs then available, evaluate up to 16 additional unseen balanced neighbors. Re-rank the complete cache.

Each recorded run used 900 main and 160 additional evaluations. The top-ten neighborhood stage refines the search; the paper pool subsequently uses the final top three per model. The stopping rules define a computational budget without certifying local or global optimality.

S1.5 Nomination and design coverage

The pool is the deduplicated union of each model's top three, four pre-existing fixed-site controls, and the empty configuration. M0 nominates P01, P02, P03; M1 nominates P04, P05, P06; M2 nominates P05, P04, P06, in those orders. The union has six screened designs and five controls. Every admitted configuration is evaluated under all three models.

Table S1: Frozen pool. Energy is rounded; exact module energy is 9.999 kWh.

ID	Buses	Modules/site	ρ	Total kWh
P01	21,45,47,48,52	32	0.25	1600
P02	32,37,45,48,52	32	0.25	1600
P03	25,32,45,48,52	32	0.25	1600
P04	51	1	0.375	10
P05	31	1	0.375	10
P06	34	1	0.375	10
P07	32,42,45,47,48	1	0.25	50
P08	32,42,45,47,48	1	0.375	50
P09	32,42,45,47,48	8	0.25	400
P10	32,42,45,47,48	32	0.25	1600
P11	None	—	—	0

Table S2: Archived search coverage before nomination. Counts include failed screens. Categories below station-count rows overlap.

Class	M0	M1	M2
Zero stations	1	1	1
One station	175	246	246
Two stations	218	287	311
Three stations	53	102	63
Four stations	67	178	184
Five stations	546	246	255
Total distinct evaluations	1060	1060	1060
Buses appearing in evaluated designs	59	59	59
Heterogeneous multi-site designs	681	591	611
Heterogeneous designs with finite score	675	284	310
Designs with 2–159 total modules	881	928	928
Of these, designs with finite score	796	519	535

A heterogeneous design has at least two different (n_i, ρ_i) pairs. Table S2 documents evaluations of intermediate capacities, two-to-four-site layouts, and heterogeneous designs. None appears among the six screened nominees. Their exclusion reflects the screening score and top-three truncation; their relative full-horizon costs remain outside the frozen-pool comparison.

The models use equal budgets, common seeds and catalogs, and union nomination. Model-dependent scores lead to different search paths. These common rules provide procedural symmetry while retaining the approximation and truncation effects of screening.

For horizon H, let $\mathcal{P}_m^{\text{complete}}(H)$ contain pool designs with a complete accepted trajectory under model m. Selection is

$$X_m^\star(H) \in \arg\min_{X \in \mathcal{P}_m^{\text{complete}}(H)} J_m(X; H). \tag{S5}$$

The quantity $g_{AB} = J_1(B) - J_1(A)$ is the pairwise cost margin between admitted configurations A and B. Selection and cost comparisons are conditional on this pool and the evaluated controller. P01 versus P10 measures location effects for their shared five-station, 32-module, $\rho = 0.25$ hardware. The comparison does not establish the relative importance of sizing and siting at other capacities, or resolve the upper catalog boundary of 32 modules per site.

S2 Physical execution and daily SOC restoration

S2.1 Hourly state propagation

M1/M2 carry SOC s, polarization u, health q, age, and Ah exposures; M2 also propagates temperature dynamically. Health is fixed during each hourly electrical integration and updated afterwards from damage. M1 imposes ambient temperature. M0 follows its constant-efficiency SOC schedule without nonlinear aging.

For a trial value of constant DC power p per cell, the one-hour integrated trajectory from the actual state must satisfy

$$0.2 \le s \le 0.9, \quad T \le 45^\circ\text{C}, \quad \frac{|I|}{C_{\text{cell}} q f_C(T)} \le 0.5 \text{ h}^{-1}, \quad |P^{\text{ac}}(p)| \le \bar{P}_i. \tag{S6}$$

Here $f_C(T)$ is the reversible capacity factor. Limits are checked on the sampled trajectory and the adaptive integration mesh. A further endpoint ceiling reserves discharge capability:

$$s_{h+1} \le c_{i,h} = \min\{0.9,\, 0.2 + \nu_i(23 - h)\}, \quad \nu_i = 0.4 \min(1, \rho_i/0.5), \quad h = 0, \ldots, 23. \tag{S7}$$

The controller applies this ceiling to reduce SOC before the final hour, when $c_{i,23} = 0.2$. Reachability is checked by integrating the actual terminal transition.

S2.2 Power correction and terminal SOC

Each station's DC target is corrected through scalar trial integrations and bracketed root finding. Network validation follows the station corrections.

1. Integrate the optimized DC target. Retain it in a non-terminal hour if all physical limits and (S7) are satisfied.

2. When correction is needed, evaluate the idle case (zero DC power). If the endpoint SOC under idling exceeds $c_{i,h}$, solve for a discharge power that reaches that ceiling. In the final hour, solve for $s_{24} = 0.2$ regardless of the optimized target.

3. Bracket the discharge root between zero and the DC power corresponding to rated AC discharge. Brent's method uses a per-cell power tolerance of 10^{-8} W. Reject the transition if rated discharge cannot reach the target, root finding fails, or the root violates a physical limit. Final-hour idle is accepted when its SOC is within 2×10^{-7} of 0.2.

4. Otherwise, if idle is feasible but the target is not, set $p = f\widehat{p}$. Find a zero of the minimum signed margin across SOC, endpoint ceiling, temperature, C-rate, and PCS power, using a bracketing interval $f \in [10^{-6}, 1]$ and a tolerance of 10^{-8}. Multiply the resulting power by $1 - 10^{-6}$ to move inside the boundary. If the margin at 10^{-6} is nonpositive, use idle. Reintegrate and independently check the accepted power. An infeasible idle transition is rejected.

The minimum margin is computed in native units and used as a feasibility-boundary root function. Every returned power undergoes an independent physical check. The procedure makes no optimality claim for the corrected dispatch.

The day is rejected if any $|s_{24} - 0.2| > 10^{-4}$. Values within 2×10^{-7} of 0.2 are snapped to 0.2 to remove numerical residue. The boundary condition is thus enforced on accepted trajectories through terminal integration and an endpoint check. Failure to find a physically feasible correction terminates the evaluation.

S2.3 Replacement and maintenance through midnight

At the beginning of each hour, M1/M2 replace a pack with $1-q \geq 0.2$. A replacement sets $q = 1$, polarization and accumulated damage/exposure counters to zero, temperature to ambient, and SOC to 0.2. Discarded old-pack energy is recorded without credit. Initializing the new pack at SOC 0.2 creates a commissioning energy requirement

$$E_i^{\text{cond}} = E_i f_C(T_h^{\text{amb}})\, 0.2/0.95. \tag{S8}$$

This energy is purchased as metered grid electricity.

The remaining-day convex dispatch is re-solved from actual post-event states. The initial trial maintenance duration is the ceiling of commissioning energy divided by PCS power remaining after idle converter loss, maximized over affected stations. Integer durations are tried in increasing order up to the remaining hours; the first verified solve is used. Longer windows are not nested feasible sets because they also reduce operating availability.

During maintenance, battery DC power is zero while commissioning demand consumes PCS and network capacity. Baseline offline reactive support is disabled. If maintenance reaches midnight, the new pack stays at its initialized SOC 0.2 under zero current; normal terminal-power restoration is bypassed for that offline pack. The same daily SOC check applies. Unpaid commissioning energy above 10^{-5} kWh at day end causes rejection. If no remaining-day window satisfies these requirements, evaluation fails under the commissioning policy.

S3 AC-network validation and lifecycle cost accounting

S3.1 Network injections after correction

Corrected DC power is converted to actual AC power using the physical converter root. In cell-watt normalization,

$$p^{\text{ac}} = \frac{2(l_0 + p^{\text{dc}})}{1 + \sqrt{1 - 4l_2(l_0 + p^{\text{dc}})}}, \qquad P_i^{\text{term}} = N_i p_i^{\text{ac}}/1000 + P_i^{\text{cond}}. \tag{S9}$$

l_0, l_2 are station-scaled converter-loss coefficients. A trial operating point is rejected if the discriminant is negative or the active-power limit is exceeded. Reactive power is clipped to the remaining PCS capability as follows:

$$Q_i^{\text{term}} = \text{clip}\left(\widehat{Q}_i, -\sqrt{\bar{S}_i^2 - (P_i^{\text{term}})^2}, +\sqrt{\bar{S}_i^2 - (P_i^{\text{term}})^2}\right). \tag{S10}$$

Offline stations use zero reactive power in the baseline. PV setpoints remain those of the current optimized plan, including any replacement-triggered replan.

An AC Newton–Raphson power-flow calculation is performed for each executed hour using the actual battery injections, PV output, and loads. It checks all bus voltages against 0.95–1.05 pu and apparent power at both ends of branches with positive published ratings. Zero or missing ratings are not finite thermal limits. Nonconvergence or a violation rejects the

evaluation; no corrective AC optimal power flow follows. Local battery correction can thus invalidate network feasibility even when the original convex schedule is feasible.

The convex model also bounds squared current by $(\bar{S}_{ij}^{pu}/1.05)^2$. This conservative representation of apparent-power limits means dispatch infeasibility does not alone prove infeasibility of a less restrictive physical AC problem.

S3.2 Operating cost and lifecycle ledger

Actual grid power comes from the AC slack-bus result, including losses and commissioning load. Operating cost is

$$C_y^{\mathrm{op}} = \sum_{d,h \in y} \left[\lambda_{d,h} P_{d,h}^{\mathrm{grid,AC}} + 0.03 \sum_j P_{j,d,h}^{\mathrm{PV}} \right] \Delta h, \qquad \Delta h = 1 \text{ h}. \tag{S11}$$

The PV term charges generated energy. Actual corrected powers enter the AC calculation used to determine purchased energy and operating cost.

Full-horizon accounting uses

$$J_m(X;H) = C^0(X) + \sum_{y=1}^{H} \frac{C_{m,y}^{\mathrm{op}} + C_{m,y}^{\mathrm{rep}} + C_y^{\mathrm{OM}}}{(1+r)^y} - \frac{V_{m,H}}{(1+r)^H}. \tag{S12}$$

Purchases occur at physical replacement events; annual cash flows are discounted at year end. Replacement commissioning is already in operating cost and is not added again. Initial commissioning enters C^0 once. The screening aging proxy and dispatch penalty are excluded from (S12), avoiding double counting with explicit purchases.

Zero terminal value sets $V_{m,H} = 0$. The health-linear alternative uses

$$V_{m,H} = \sum_i E_i c_E \max\left\{ \frac{q_{i,H} - 0.8}{0.2}, 0 \right\}, \qquad c_E = 1620 \text{ CNY/kWh}. \tag{S13}$$

Terminal recovery uses health at the chronological endpoint. Annual inputs repeat over 15 years while aging states continue across years. Financial views use completed 10-, 12-, and 15-year prefixes. A failed horizon has no complete NPC.

S4 Supplementary evaluation records

S4.1 Chronological completion status

Of the 33 model–configuration evaluations, 29 complete 15 years. Table S3 lists the completed counts and the first recorded stopping points. P04 and P05 complete under all three models; P06 completes under M0 and M2.

Table S3: Completion of the frozen-pool evaluations over 15 years.

Model	Completed / admitted	Evaluations stopping before 15 years
M0	10 / 11	P11: year 1, weather day 158
M1	9 / 11	P11: year 1, weather day 158; P06: year 10, weather day 209
M2	10 / 11	P11: year 1, weather day 158

P06/M1 stops after 3493 completed days and therefore has no complete 10-year cost. The P11 outcome records failure of the specified rolling-dispatch procedure. The available record does not identify a binding bus/hour/constraint or establish a minimum physical storage requirement. Consequently, the cost comparison uses completed storage trajectories and provides no payback estimate relative to a complete no-storage trajectory.

S4.2 Selected-day integration checks

Thirteen matched standard/strict replays cover P01/P05 winter and summer extreme days, the first P01 replacement events at buses 21 and 47 under M1/M2, and the corresponding M1s bus-21 event. Each replay starts from the lifecycle state for that day. All 13 pairs complete without network or temperature violations. Replacement buses, hours, and purchase amounts agree between the two integration settings.

Table S4: Nominal selected-day integration checks. Differences compare the standard and strict replay of the same day and initial state.

Metric	Result
Matched replay pairs completed	13 / 13
Replacement bus/hour/purchase mismatches	0
Maximum daily operating-cost difference (CNY)	2.24×10^{-4}
Maximum end-day SOH difference	2.34×10^{-10}

These checks quantify numerical agreement on the selected days. They do not establish strict-integration accuracy over every day of the full lifecycle.

S5 Numerical settings

Standard execution uses LSODA with a relative tolerance of 10^{-8}, a maximum step size of 300 s, and a 10-s stress-quadrature interval. Absolute tolerances are 10^{-7} for temperature, 10^{-9} for SOC, and 10^{-10} for polarization. Selected strict replays use Radau with a maximum step size of 30 s and a 1-s quadrature interval. They do not constitute a full-lifecycle strict-integration rerun.

Physical checks allow 2×10^{-7} for SOC and endpoint ceiling, 10^{-6} °C for temperature, 2×10^{-6} h^{-1} for C-rate, and 10^{-5} kW for station active power. The AC mismatch tolerance is 10^{-8} MVA; voltage and normalized loading checks allow 10^{-5}. Daily SOC rejection separately uses 10^{-4}.

References

[1] Y. Li, Y. Gu, G. He, and Q. Chen, Optimal dispatch of battery energy storage in distribution network considering electrothermal-aging coupling, *IEEE Transactions on Smart Grid*, vol. 14, no. 5, pp. 3744–3758, 2023, doi: 10.1109/TSG.2023.3243021.